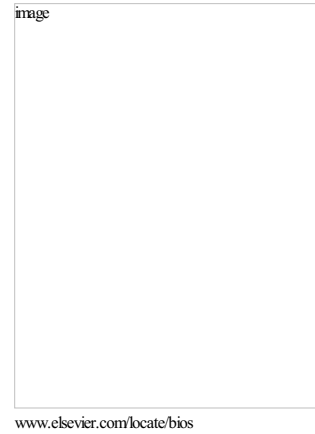


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Imaging Evidence of the Effect of Socio-Economic Status on Brain Structure and**Development**

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Abstract

Numerous studies have shown an association between children's socio-economic status (SES) and disparities in neurocognitive development, achievements and function later in life. Research focus has recently shifted to imaging of the brain's response to the child's environment. This review summarizes the emerging studies on the influences of early-life SES on brain structure and development, and addresses the relation between brain development and enriched environments. The studies provide evidence of significant associations between SES and brain structure, growth and maturation, not only in healthy infants and children but also in infants with medical conditions. This suggests that the relation between SES and later-life function and achievements operates through alterations in brain maturation. While the brain changes seem to persist without intervention, animal models of environmental enrichment show the potential of SES-related brain changes to be reversible and dynamic. This review underscores the critical need for reducing the impact of socio-economic disparities and early targeted and prolonged interventions, and highlights the potential of these interventions leading to optimal opportunities for our youngest.

Introduction

The importance of early socio-economic exposures for health and educational developmental trajectories is increasingly apparent. The cost to society of inaction to promote early child health and development is staggering. As an example, in Canada, the costs of adolescents not completing school is estimated by the Canadian Council on Learning to be CAD 2.6 billion annually for each cohort of early school leavers. This amounts to a CAD 17.6 billion lifetime cost for each cohort in revenue loss in taxes and employment insurance premiums (1). Early child development not only impacts health in children but also effects long-term health, cognition and achievement (2).

The human brain undergoes an impressive phase of development from fetal life through childhood (*Figure 1*). This period of dramatic brain development is accompanied by a heightened vulnerability of the brain. Development may be influenced directly through injury by events such as stroke or trauma. It may be influenced indirectly by systemic conditions such as heart disease or cancer and its treatment. There is also increasing recognition that the brain is remarkably sensitive to the child's environmental circumstances during this period, such as socio-economic status (SES), maternal depression, social deprivation, nutrition and growth.

Differences in children's SES, generally measured by one or both of their parents' SES, has been found to be associated with disparities in neurocognitive development and, later in life, with functional outcomes such as language skills and academic achievements. Consistent across studies is the observation at a group level that lower SES predicts less robust developmental trajectories (3-7). More recently, this issue has been explored using rigorous techniques, including newer brain imaging modalities, which provide an unprecedented window of the brain's response to the child's environment, whether adverse or supportive. Recent studies have demonstrated a link between SES and brain development in healthy infants and children (7-20), implicating that the relation between

SES and later-life function may be mediated by alterations in brain maturation. MR imaging now has the potential to serve as a biomarker for intervention and provide an early window to improve interventions.

While the brain's vulnerability during the earliest years increases its susceptibility to ill-effects of SES, research based on animal models has shown a remarkable capacity for brain recovery and repair. Animal models demonstrate that environmental enrichment leads to improved brain development and function. Such research suggests the potential in humans to promote optimal outcomes despite early-life adversity, whether medical (e.g. epilepsy) or environmental (e.g. toxic stress).

In this review, we focus on the relation between SES and brain structure and development. We summarize the emerging studies on the influences of early-life SES on brain structure and development in healthy infants. Subsequently, we address the relation between brain development and enriched environments, and the potential of SES-related brain changes to be reversible and dynamic. Finally, we summarize the SES-related brain changes and their possible reversibility, and highlight the potential of improved early and targeted interventions leading to optimal opportunities for our youngest.

Influences of Socio-Economic Status on Changes in Brain Structure and Development in Healthy Infants

SES is a multidimensional construct that refers to an individual's access to economic and social resources, as well as the benefits and social standing that come from these resources. The notion of SES is distinct from the notion of poverty. Poverty implies a threshold that dichotomizes children and families as being poor, or not. SES, on the other hand, describes a more 'continuous' notion of the

extent of resources held by children and families. As such, it is typically measured by parental income or income-to-needs, parental education and parental occupation. While a rich literature has demonstrated the association between SES and a wide variety of health and developmental outcomes, from adverse birth outcomes to infant mortality, in this review we specifically focus on the association between SES and neurodevelopment, examined via brain imaging studies.

While earlier studies of SES and brain structure and development were mostly directed at older infants and adolescents up to adulthood (8-18), more recent studies focus on links between SES and the brain at younger ages (19-20). Table 1 summarizes the studies described below including their study population, design, SES measure, outcome measures and main findings. Although the studies differ with respect to study design and population studied, they all address the relation between SES and brain structure and development using MRI. In these studies, SES is defined as income or income-to-needs and/or parental education.

Child and Youth MRI studies: brain structure

Most studies in infants and adolescents on SES and the brain focused on brain regional volumes (8-13, 20). The earliest study by Raizada et al. (8) describes a small cohort of children of diverse SES undergoing (functional) MRI at age 5 years. Correlations were found between SES and gray and white matter volumes of the inferior frontal gyrus, with lower SES children having smaller volumes. SES also predicted hemispheric specialization of the left inferior frontal gyrus during a rhyming task, being related to reading skills; this brain region contains Broca's area associated with speech and language function. These findings are of interest given the previously described links of SES and language performance in older children. In another study (20), hippocampal gray matter volume was assessed in 317 children (aged 4-18 years) across the socio-economic spectrum. The group of children from lower income backgrounds had lower hippocampal volumes, a brain region involved in learning and

memory and known to be “stress-sensitive”. In addition, in two studies by Noble et al. on the same cohort (9, 12), 60 socio-economically diverse children were evaluated at a mean age of 11.4 years. After correcting for age, total cortical volume and sex, higher income-to-needs was associated with larger hippocampal volume and lower parental education with larger amygdala volumes, whereas a lower SES composite score was associated with smaller amygdala volumes.

Following on these initial studies assessing individual stress-sensitive brain regions, research interest expanded to other brain regions. Yet, cross-sectional MRI data from 325 children (aged 4.5-18 years) in the National Institutes of Health Magnetic Resonance Imaging Study of Normal Brain Development, showed no relations between family income and volumes of several different brain regions (10); See Table 1 for details. Of interest, these brain regions studied were not previously related to the human stress response. Furthermore, this large study assessed only normally developing children.

The concept of a *maturation lag* of the brain is increasingly attributed to exposure to low SES. In the study by Jednorog et al. (15), several brain regions were assessed in healthy 10-year-old children varying in SES. Lower SES predicted smaller gray matter volumes of hippocampi and middle temporal, left fusiform and right inferior occipito-temporal gyri. Interestingly, they also identified local gyrification effects in anterior frontal regions in relation with lower SES, suggestive of a potential brain developmental lag in lower SES children. Luby et al. (11) found early exposure to a low income-to-needs ratio to be associated with smaller cortical gray matter, white matter, left hippocampal and left amygdala volumes in children aged 6–12 years. After correction for SES associated factors such as caregiver education, supportive/hostile parenting, children’s experience of stressful life events, the relation remained significant for white and gray matter volumes. Importantly, hippocampal volume was fully mediated by caregiver behavior. In a longitudinal cohort study, Hair et al. (13) analyzed repeated MRI scans (3 consecutive years) of 301 participants aged 4-

22 years for relation between family income and regional gray matter volumes. Again, low income was associated with smaller gray matter volumes, being more prominent with lower income. Of interest, 20% of the difference in cognitive and academic achievement assessments could be explained by maturational lags in the frontal and temporal lobes.

SES has also consistently been related to lower cortical thickness. Lawson et al. (16), studying 283 healthy children and adolescents aged 4-18 years, found lower parental education to predict decreased cortical thickness in anterior cingulate and superior frontal gyrus (prefrontal cortex). In line with this, a recent large cohort study (n= 1,099) in children and adolescents aged 3-20 years (18) showed increasing income (not education) to be logarithmically associated with increasing brain surface area. Again, the association was strongest at the lower end of the SES spectrum. Together, these data highlight the important link between SES and cortical development, in addition to the subcortical changes (e.g. hippocampus) described in earlier studies.

Infancy MRI studies: brain structure

Recent MR imaging studies in infants reveal that the brain undergoes dramatic development in the first year of life. This dynamic state of the brain is accompanied by vulnerability that is critical given the importance of early brain growth for later-life function (21-26). Perhaps not surprisingly, there is a growing interest in the SES-related brain changes in younger infants and in the onset of the changes (17, 19). In another longitudinal study (17), growth trajectories of deep gray matter were assessed from 203 scans of 77 children (aged 5 months-4 years). Low income children showed significantly slower growth rates as compared to high income children from 5 months of age onwards. Finally, Betancourt et al. (19) studied a group of 44 healthy, single-race female infants from varying SES backgrounds. They showed smaller cortical and deep gray matter volumes in infants of lower SES than in their wealthier counterparts, even as early as 5 weeks of life.

Related to the finding that changes in brain structure and development are already present within the first few weeks after birth (19), other studies have shown that children from lower income groups have persistently smaller head circumference measurements than those from higher income groups (27-28). Head circumference has variably been described to reflect brain growth and development, especially in early childhood (29-30). Differences in neurobehavioral development between different social classes have already been described in fetuses from mid-pregnancy onwards (31). In addition, the influences of SES seem to become stronger with increasing exposure to less favorable SES factors (8, 13, 18). Differences in achievement between different income groups have been reported to emerge early and widen throughout later years (32). Except for some inconsistency between studies with regard to hippocampal volumes, similar differences in brain volumes have been described between adults from different income groups (33-34). Together these data suggest that SES relates to brain changes from early in life, yet the full impact of these brain changes are only measured later in the life-course.

To date, most studies in this area are cross-sectional with only two being longitudinal. Despite this, the data are congruent in demonstrating a relation between SES and brain growth and development. This particularly applies to hippocampal, amygdala, gray matter and white matter volumes and cortical development. Importantly, the relation is evident from a very young age. Furthermore, the available longitudinal data support the hypothesis that the link between SES and later-life function is mediated by alterations in brain maturation. However, our understanding of the complex links between SES and the brain and their implications are incomplete. Important questions still need to be answered. Are the brain changes permanent or reversible? If changes are reversible, what is the most effective window for intervention? Are certain brain areas more vulnerable than others and at different periods in development? Does SES only affect healthy infants, or also infants who are ill or have brain injury?

The structural brain imaging studies are complemented by emerging studies focusing on the relation between SES and brain function, using functional MRI. The rate of functional network development in the brain is the greatest in the first year of life, going hand in hand with fast synaptogenesis, dendrite elaboration, myelination, white matter tract development and continued subplate growth (7, 35-42). Gao et al. (42) examined the relative maturation sequence of 9 key brain functional networks and their relation with SES in normally developing infants using longitudinal resting-state functional MRI during the first year. They found marginally significant positive SES-brain correlations for both the sensorimotor and default-mode networks, being most salient at 6 months of age. They also identified a maturation sequence from primary sensorimotor/auditory to visual to attention/default-mode and finally to executive control networks, together with network specific critical growth periods. In this study, higher SES predicted more rapid functional network maturation.

Furthermore, the structural brain changes detected with MRI seem to matter in regards to functional outcomes. In a recent study discussed above (13), structural brain changes accounted for 20% of the testing gap between poor and middle class or well-off children. A study in adolescents showed that cortical thickness in all lobes of the brain was greater in students from higher than lower income backgrounds. In addition, greater cortical thickness, particularly in temporal and occipital lobes, was associated with better test performance (43).

SES, Brain Injury, and Brain Development

SES is also related to brain growth in children at increased risk of brain injury and/or abnormal development. In a study of term born infants with presumed hypoxic-ischemic encephalopathy, infants followed from 12 to 30 months of age had a greater risk of deterioration in

neurodevelopmental outcome scores over this interval if in lower income homes (44). Stiver et al. (45) described an association between parental education and cerebellar volume in preterm infants. Maternal postsecondary education was significantly related to term and 2-year cerebellar volumes, with higher education going along with greater volumes by 2 years. This indicates that maternal education is associated with improved growth detectable by 2 years, and thus positive influences on brain development. Interestingly, cerebellar volumes were not associated with SES in healthy infants (*Table 1*). Previously, associations have been described between low SES and increased risk of preterm birth, intra-uterine growth restriction and hypoxic-ischemic encephalopathy, and between lower SES and adverse neurodevelopmental outcome in high-risk infants (28, 46-49). In addition, infants born preterm and/or growth restricted and those with hypoxic-ischemic encephalopathy are prone to brain injury and/or smaller brain volumes, and are at increased risk of adverse neurodevelopmental outcome (50-53).

Of interest, the brain regions being most affected by preterm birth and intra-uterine growth restriction overlap with those affected by SES. Over recent years, neuro-imaging studies in high-risk neonates have shifted from brain injury to dysmaturation, making us more aware of the importance of altered brain maturation for long-term outcome (41). These data suggest that unfavorable SES factors may place high-risk infants at even greater risk of less optimal brain development. In addition, SES-related factors may be important mediators of longer-term brain repair in these infants.

The Potential to Enhance Brain Development

Brain imaging studies suggest that SES-related factors influence brain structure and development. Studies also suggest that the developing brain's capacity for recovery and repair provides opportunities for interventions to dampen or even reverse the ill-effects of early adversity, whether environmental or medical.

One of the oldest and often cited early childhood intervention programs with long-term follow-up is the Carolina Abecedarian (ABC) Project (2). The randomized intervention was divided into a first stage of early intervention, focusing on development of language, emotional regulation and cognitive skills, and a second stage, focusing on improvement of early mathematics and reading skills. The intervention group had consistently better developmental and education achievements. Besides, children in less favorable conditions randomly assigned to treatment were shown to have significantly lower prevalence of risk factors for cardiovascular metabolic disease in their mid-30s. The Perry Preschool Program randomized low income 3- and 4-year-old infants to either a high-quality preschool program or no preschool. Infants receiving preschool demonstrated positive and sustained impacts on achievement test scores, educational attainment and social skills but not on IQ (54). Non-invasive brain imaging technologies, such as MRI, now provide an opportunity to assess the brain changes resulting from these interventions, and ultimately to enhance interventions for most robust impact.

The beneficial effects of these early child health intervention programs on later function are promising and suggest that the impact of environmental factors on later function may be preventable and even reversible. Robust neuroscience studies in animal models have shown that rearing environments affect brain development and function. Animals raised in enriched environments, including toys, social stimulation, exercise and novelty, show positive changes in brain structure and function compared to their less favorable counterparts. These beneficial changes include increases in cortical gray matter and hippocampal volumes, enhanced hippocampal neurogenesis and integration of newborn cells into functional circuits, down-regulated stress response in amygdala, and alterations in neurotransmitter and neurotropic systems (55-61). Environmentally enriched animals also demonstrate better learning and memory skills, great capacity for plasticity and behavioral adaptation, and reduced anxiety- and depression-related behavior (58-

60). Besides, several studies have shown that enriched environments can reverse deficits in models of brain dysfunction, including brain damage, dementia and aging (62-66).

In a recent study by Leger et al. (58), the beneficial effects on different behavioral tests and neurobiological parameters, such as neurogenesis and stress-related hormone levels, was assessed after different durations of enriched environment exposure (24 hours to 5 weeks). They showed that even a 3-week exposure was sufficient to increase serotonin levels in the frontal cortex and enhance hippocampal cell survival. However, the brain changes did not persist after cessation of enrichment. Thus, duration of exposure is a key determinant of the beneficial brain health effects of enriched environment.

The brain regions described in these animal models showing recovery and increased levels of neurotransmitters in enriched conditions are regions involved in cognitive processes such as learning, memory, fear, emotional processing and executive function. These regions largely overlap with the brain regions linked to SES in human studies. Developing brain biomarkers to facilitate the evaluation of interventions is a key opportunity for future research to enhance the implementation of strategies to promote optimal brain health outcomes. Advanced MRI techniques, including volumetric MRI, diffusion tensor imaging and functional MRI, are promising in this regard.

Conclusions and Future Perspectives

Studies on the link between SES and neurocognitive development, achievements and function later in life are numerous, with the strongest relationships described in language skills, learning, memory, executive functions and regulation of cognition and emotions. Despite differences across brain imaging studies and their apparent limitations, these studies provide evidence of significant and quantifiable associations between SES and brain structure, growth and maturation in healthy infants

and children. The most prominent relations are described for stress-sensitive regions that have important roles in memory, decision-making, language and other aspects of emotion and intelligence. The findings of these studies add to the previously described relation between SES and later-life function and achievement, and indicate that this relation operates through alterations in brain maturation: the link between SES and the brain is detected at a very early age, and without intervention, seems to persist into adulthood. Additional longitudinal studies, and studies that consider the impacts of interventions, are certainly needed. SES also seems to modify brain maturation in infants with medical conditions such as preterm birth and brain injury. Understanding this relation better may provide new avenues to promote brain repair in the setting of brain injury. Studies in twins have shown strong genetic effects on neonatal MRI outcome (67). Brain imaging may also enhance our capacity to determine the genetic contribution to a child's sensitivity to his or her environment.

The long-term impact of SES-related brain changes is a critical issue for future study as 40% of the children born in the lowest income families will remain in the lowest income groups as adults. Low income is not only a limited supply of money but goes hand in hand with differences in nutrition, prenatal care, parental education level and other environmental stressors. These findings highlight that current programs making efforts to improve child health and development may intervene too late. In addition, they may not be continued long enough to maintain the benefits gained with early intervention and/or targeted at the right individuals and/or environmental factors. This is a call for action, and underscores the critical need for early targeted and prolonged interventions for our youngest.

Despite the complex and multifactorial links between SES and later neurocognitive function, the brain imaging studies reviewed here point to a link between SES and brain maturation. The findings underscore that with the transformative improvements we have seen in pediatric medicine,

attention is still required to ensure that not only the sickest of the sick but also healthy infants with less fortunate environmental circumstances develop optimally. There are no data to suggest that the effects of SES are irreversible or deleterious, and it is the responsibility of the scientific community not to stigmatize infants living in less favorable SES conditions; many infants are thriving despite limited socio-economic resources. The existence of the SES-related brain changes early in life suggests that interventions cannot begin too early in support of young children and their families, and that programs and policies directed at reducing the impact of socio-economic disparities are essential.

Table and Figure

Table 1.

Studies describing associations between SES and brain structure and development. Study characteristics described include study population, design, definition of SES measure, MRI outcome measures, and main findings in relation to lower SES measure. The studies are ordered by study design, longitudinal followed by cross-sectional studies, and subsequently by age range of the study population, with the lowest age range reported first.

(SES, socio-economic status; wk, week; mo, month; y, year; N, number of infants; ITN, income-to-needs).

Figure 1.

T₁-weighted images acquired in a very preterm born neonate within a few weeks of birth (left) at 28 weeks' postmenstrual age, at 40 weeks' postmenstrual age or term equivalent age (middle), and at 6 years of age (right). Figure courtesy of Dr. Emma Duerden and Dr. Margot Taylor, The Hospital for Sick Children.

Study	Study population Age range (mean) Cohort size	Study design	SES measure	MRI outcome measures	Results Lower SES measure associated with:
Hanson et al., 2013	5 months–4 years (13.5 mo) N = 77	Longitudinal	Family income	Volumes: - White matter - Cortical and deep gray matter	Smaller gray matter volumes (frontal and parietal)
Hair et al., 2015	4–22 years (12 y) N = 389	Longitudinal	Family income	Gray matter volumes: - Total brain - Frontal and temporal lobes - Hippocampus	Smaller regional volumes (note: effects strongest with lower SES measure)
Betancourt et al., 2015	4–6 weeks (5 wk) N = 44	Cross-sectional	Composite score: - ITN - Maternal education	Volumes: - White matter - Cortical and deep gray matter	Smaller gray matter volumes
Noble et al., 2015	3–20 years (11.9 y) N = 1099	Cross-sectional	Composite score: - Family income - Parental education	Volume and thickness of cortical surface area	Smaller volumes (note: effects strongest with lower SES measure)
Hanson et al., 2011	4–18 years (11.2 y) N = 317	Cross-sectional	Composite score: - Family income - Parental education	Volumes: - Total brain - Hippocampi and amygdala	Smaller hippocampal volume
BDCG et al., 2012	4–18 years (10.9 y) N = 325	Cross-sectional	Composite score: - Family income - Parental education	Volumes: - Total brain - Deep gray matter - Frontal, temporal, parietal and occipital lobes - Lateral ventricles - Cerebellum - Brainstem	No association
Lawson et al., 2013	4.5–18.3 years (11.5 y) N = 283	Cross-sectional	Composite score: - ITN - Parental education	Cortical gray matter thickness	Lower thickness right anterior cingulate and left superior frontal gyrus
Raizada et al., 2008	5–5.7 years (5.3 y) N = 14	Cross-sectional	Composite score: - Parental education - Occupation	Gray and white matter volume of inferior frontal gyrus	Smaller volumes (note: effects strongest with lower SES measure)
Noble et al., 2012	5–17 years (11.4 y) N = 60	Cross-sectional	Composite score: - ITN	Volumes: - Left temporal and fusiform gyri	- Smaller amygdala volume with parental

			- Parental education	- Hippocampi and amygdala - Anterior cingulate cortex	education - Smaller hippocampal volume with ITN
Noble et al., 2013	5–17 years (11.4 y) N = 60	Cross-sectional	Composite score: - ITN - Parental education	Volume: - Hippocampus and amygdala - Anterior cingulate - 5 language areas	- Smaller hippocampal volume - Larger amygdala volume
Luby et al., 2013	6–12 years (9.8 y) N = 145	Cross-sectional	ITN	Volume: - White matter - Gray matter - Hippocampus and amygdala	Smaller volumes
Jednorog et al., 2012	8–10 years (9.6 y) N = 23	Cross-sectional	Composite score: - Maternal education - Occupation	Volume: - Total brain - Hippocampi - Middle temporal, left fusiform and right and inferior occipito-temporal gyri White matter microstructure	Smaller gray matter volumes: hippocampi, parahippocampal region, middle temporal gyrus, insula, left fusiform gyrus, right inferior occipito-temporal region, left superior middle frontal gyrus

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Brain Development: from 28 weeks gestation to 6 years of age

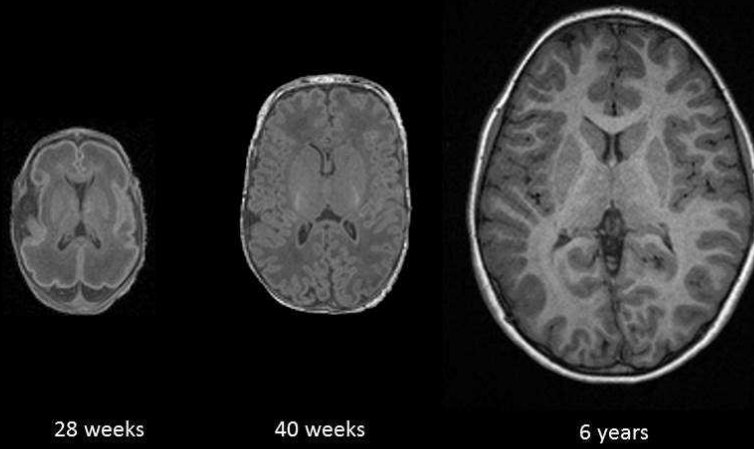


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